Logistics and Fleet Management Data Analytics Project

1. Project Overview & Objectives

The primary objective of this project is to analyze and optimize logistics and fleet management operations. By leveraging raw operational data, the project aims to build an end-to-end data pipeline that cleans historical records, segments trip performance using Machine Learning, and deploys interactive Business Intelligence dashboards to track operational efficiency, financial performance, and fleet utilization.

2. Data Description

The dataset consists of a comprehensive logistics database containing multiple interrelated tables. Key tables encompass:

Customer & Driver Data: customers, drivers, driver_monthly_metrics.

Fleet & Logistics Operations: trucks, trailers, loads, trips, delivery_events, routes.

Operational Costs & Safety: fuel_purchases, maintenance_records, safety_incidents, facilities, truck_utilization_metrics.

3. Data Cleaning & Preprocessing

Extensive data cleaning and Exploratory Data Analysis (EDA) were conducted using Python (Pandas) to ensure data integrity before analysis:

Data Extraction & Profiling: Extracted 15 datasets from a centralized archive (data.zip) and performed initial profiling to check dimensions, null values, and data types.

Statistical Analysis: Calculated skewness and detected numerical outliers using the Interquartile Range (IQR) method.

Data Standardization: Stripped leading/trailing whitespaces from categorical text columns and correctly formatted date columns into datetime objects for time-series compatibility.

Missing Values Handling: Filled missing reference IDs (e.g., truck_id and driver_id in fuel purchases) to maintain relational integrity.

Exporting Clean Data: Exported the processed and verified datasets to a dedicated data cleaned directory for seamless integration with downstream models and BI tools.

4. Machine Learning & Predictive Modeling

A Machine Learning clustering model was developed to segment and analyze trip profiles:

- **Feature Engineering & Merging:** Merged the trips, trucks, and loads tables to create a unified dataset containing critical features: actual_distance_miles, actual_duration_hours, idle_time_hours, and weight_lbs.

- **Data Scaling:** Applied StandardScaler to normalize the numerical features, ensuring uniform distance calculations for the clustering algorithm.

- **Clustering (K-Means):** Trained a K-Means algorithm to group trips into 3 distinct clusters (n_clusters=3), achieving a silhouette score that validates the cluster quality.

- **Model Deployment Preparation:** Saved the trained K-Means model (kmeans.pkl) and the scaler (scaler.pkl) using the joblib library, allowing the model to accurately predict and segment new incoming trip data.

5. Data Visualization & BI Deployment

Three specialized Power BI dashboards were developed utilizing advanced DAX measures to translate raw data into actionable business metrics:

1. **Operation Dashboard:** Focuses on daily logistical execution.

    - **Key Measures:** Average Idle Hours, Average Load Weight, Miles per Truck, and On-Time Delivery (OTD) rate.

    - **Visuals:** Clustered bar charts for load counts, geographical mapping of detention minutes by city, and trend lines for maintenance costs versus load weights.

2. **Strategic Dashboard:** Focuses on financial health and profitability.

    - **Key Measures:** Total Revenue, Gross Profit Margin, Profit per Load, Revenue per Load, and Top Customer Revenue %.

    - **Visuals:** Revenue funnels by booking type, map visuals sizing Gross Profit Margin by destination state, and breakdown of revenue per customer.

3. **Efficiency Dashboard:** Focuses on cost control and asset utilization.

    - **Key Measures:** Cost Per Mile (CPM), Fuel Cost Per Mile, Miles Per Gallon (MPG), and Total Maintenance/Fuel Costs.

Visuals: Tables tracking maintenance costs per truck ID, donut charts categorizing maintenance by truck make, and stacked columns for fuel costs across locations.

6. Extracted Insights

Trip Segmentation Insight: The clustering model successfully identified different trip behaviors (e.g., short-haul vs. long-haul) which correlate directly with varying levels of idle_time_hours and actual_duration_hours.

Financial Insight: Analyzing Total Costs (Fuel + Maintenance) against Total Revenue allowed for the precise calculation of Profit per Load, highlighting which booking types and customer contracts yield the highest Gross Profit Margin.

Operational Efficiency Insight: Tracking Cost Per Mile (CPM) and Miles Per Gallon (MPG) across different truck makes provides a clear view of fleet depreciation and fuel efficiency.

7. Recommendations

Fleet Optimization: Utilize the ML trip clusters to assign the most fuel-efficient trucks to high-duration/long-haul segments, aiming to lower overall idle hours and fuel consumption.

Cost Management: Trucks identified in the Efficiency Dashboard with consistently high Cost Per Mile (CPM) or frequent maintenance events should be evaluated for replacement or preventive servicing.

Strategic Growth: Focus business development and sales efforts on the top-performing customer segments that drive the highest Revenue per Load and optimal Gross Profit Margins.

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